Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim:Perform FFT and IFFT on discrete time signal.**	
Experiment:-5	**Date:27-09-2026**	**Enrollment No:92400133063**

Code:-

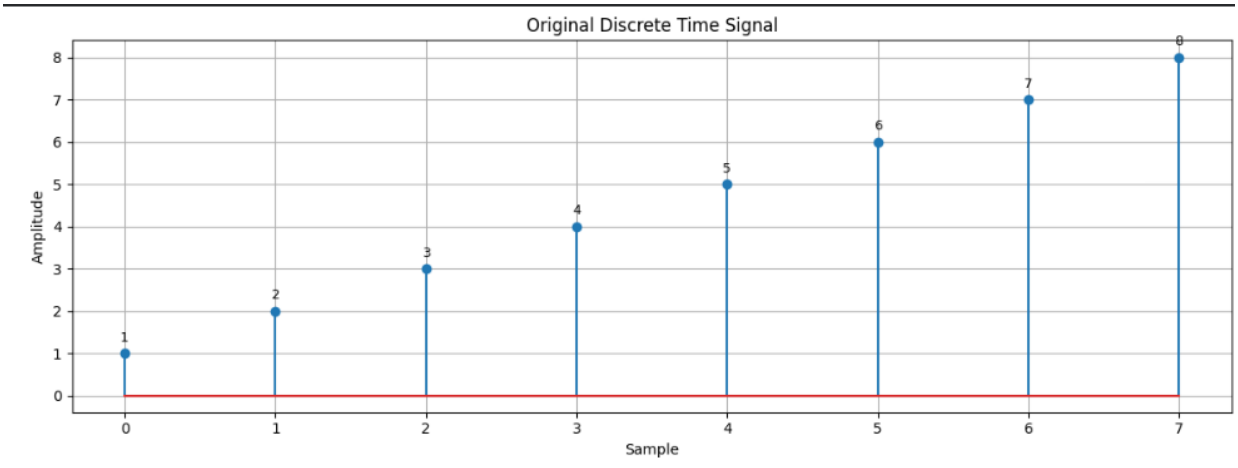
```
import numpy as np
import matplotlib.pyplot as plt
signal = np.array([1, 2, 3, 4, 5, 6, 7, 8])
# Compute the FFT of the signal
fft_result = np.fft.fft(signal)
# Compute the magnitude and Phase spectrum of the FFT result
magnitude_spectrum = np .abs(fft_result)
Phase_spectrum = np.angle(fft_result)
# Compute the IFFT of the FFT result
reconstructed_signal = np.fft.ifft(fft_result)
plt.figure(figsize=(12, 22)) # Adjust figure height for 5 subplots

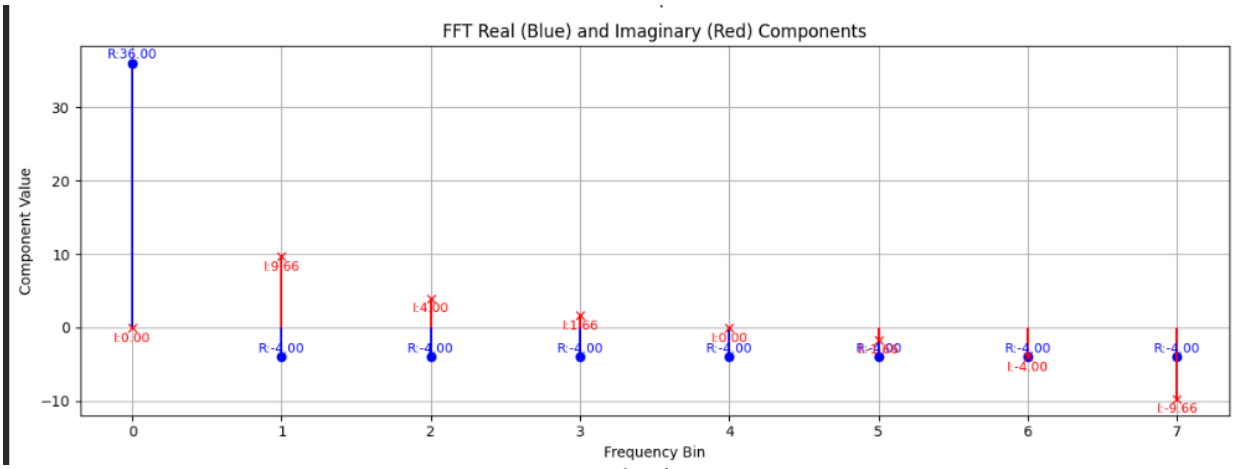
# 1. Original Discrete Time Signal
plt.subplot(5, 1, 1) # 5 rows, 1 column, 1st plot
plt.stem(signal)
plt.title('Original Discrete Time Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.grid(True)
for i, val in enumerate(signal):
    plt.text(i, val + 0.2, f'{val}', ha='center', va='bottom', fontsize=9)
# 2. FFT Real and Imaginary Components
plt.subplot(5, 1, 2)
plt.stem(fft_result.real, linefmt='b-', markerfmt='bo', basefmt=' ')
plt.stem(fft_result.imag, linefmt='r-', markerfmt='rx', basefmt=' ')
plt.title('FFT Real (Blue) and Imaginary (Red) Components')
plt.xlabel('Frequency Bin')
plt.ylabel('Component Value')
plt.grid(True)
for i, val in enumerate(fft_result.real):
   plt.text(i, val + 0.2, f'R:{val:.2f}', ha='center', va='bottom', fontsize=9, color='blue')
for i, val in enumerate(fft_result.imag):
    plt.text(i, val - 0.5, f'I:{val:.2f}', ha='center', va='top', fontsize=9, color='red')
# 3. FFT Magnitude
plt.subplot(5, 1, 3)
plt.stem(magnitude_spectrum)
plt.title('FFT Magnitude')
```

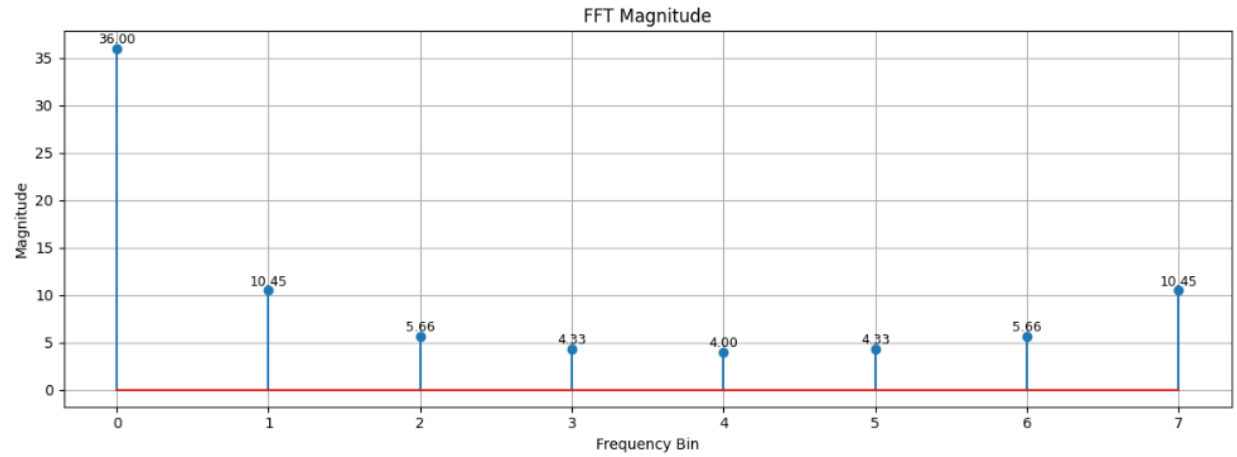
```
plt.xlabel('Frequency Bin')
plt.ylabel('Magnitude')
plt.grid(True)
for i, val in enumerate(magnitude_spectrum):
    plt.text(i, val + 0.2, f'{val:.2f}', ha='center', va='bottom', fontsize=9)
# 4. FFT Phase Spectrum
plt.subplot(5, 1, 4)
plt.stem(Phase_spectrum)
plt.title('FFT Phase Spectrum')
plt.xlabel('Frequency Bin')
plt.ylabel('Phase (radians)')
plt.grid(True)
for i, val in enumerate(Phase_spectrum):
    plt.text(i, val + 0.1, f'{val:.2f}', ha='center', va='bottom', fontsize=9)
# 5. Reconstructed Signal
plt.subplot(5, 1, 5)
plt.stem(reconstructed_signal.real)
plt.title('Reconstructed Signal')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.grid(True)
for i, val in enumerate(reconstructed_signal.real):
    plt.text(i, val + 0.2, f'{val:.2f}', ha='center', va='bottom', fontsize=9)
plt.tight_layout()
plt.show()
```

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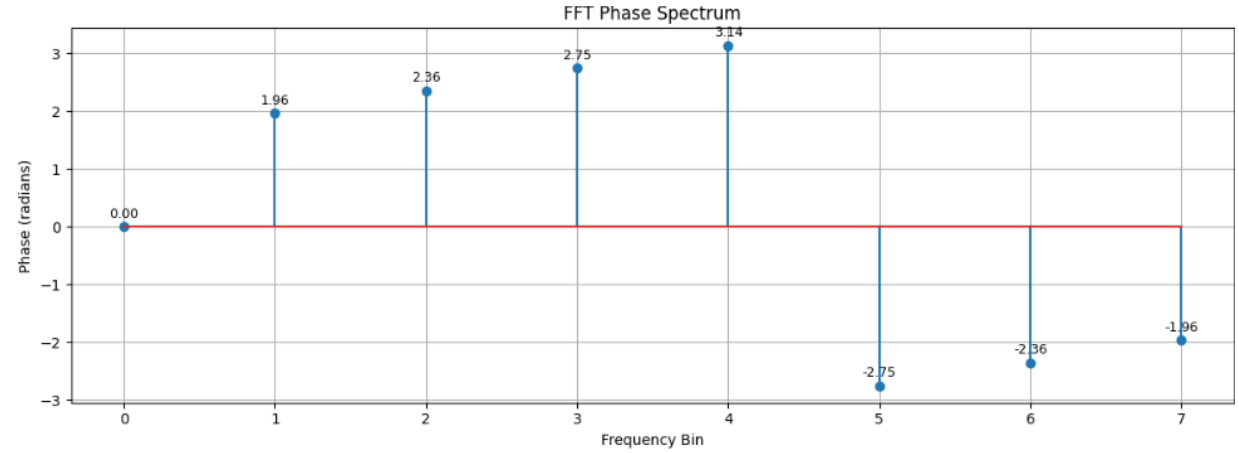
Output:-

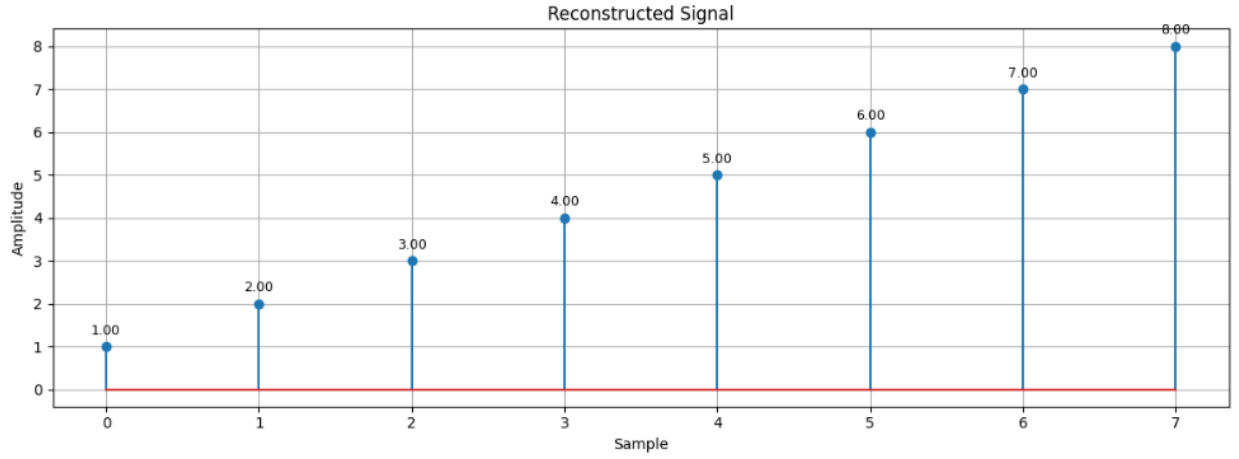






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Conclusion:-

The images illustrate the complete process of analyzing a discrete time-domain signal using the Fast Fourier Transform (FFT) and recovering it via the Inverse FFT (IFFT). The original signal, a linear ramp from 1 to 8, is decomposed into its frequency components, showing a dominant DC offset (Bin 0) and conjugate symmetry in its complex values. The magnitude and phase spectra detail the signal's frequency strength and angular shifts, respectively. Finally, the reconstructed signal perfectly matches the original, demonstrating that the FFT is a lossless and fully reversible operation.